

MODELING INTERACTION AND MULTIMODALITY IN PEDESTRIAN TRAJECTORY FORECASTING

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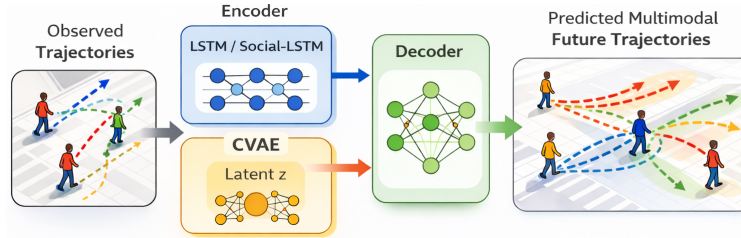


Figure 1: Pipeline for multimodal pedestrian trajectory forecasting

1 INTRODUCTION

Predicting the future motion of pedestrians and surrounding vehicles is a core component of self-driving systems because it directly affects safety and planning decisions. Given a short history of where each agent has been (and optionally some map information), our goal is to predict a distribution over plausible future trajectories for one or more target agents over the next few seconds. This problem is challenging because the future is inherently multi-modal. For example, a pedestrian at an intersection could stop, cross the street, or turn back. A vehicle might accelerate, yield, or change lanes depending on nearby traffic.

This project is meaningful because trajectory forecasting remains an open challenge in realistic mixed-traffic environments, especially when many agents interact in dense scenes. Our plan is to begin with interaction-based and non-interaction-based baseline models and then introduce a focused improvement, such as implementing Conditional Variational Autoencoders (CVAEs) to output multimodal trajectories and analyze how interaction-based modeling, multimodal trajectories, and the combination of the two approaches affect performance. Concretely, the model takes as input sequences of observed 2D pedestrian coordinates and outputs predicted future (x, y) positions for each agent. Models are trained with Gaussian NLL loss to learn a probabilistic trajectory distribution, using the ETH/UCY benchmark datasets.

2 RELATED WORK

Early deep learning approaches for pedestrian trajectory prediction relied primarily on recurrent architectures to extrapolate motion, with Social-LSTM (Robicquet et al., 2016) introducing social pooling to model agent interactions and mitigate collisions. While foundational, recurrent models often struggle with cross-dataset transferability, a challenge addressed by the Spatial-Temporal-Spectral (STS) LSTM (Zhang et al., 2020), which utilizes discrete Fourier transforms to capture generic, frequency-based motion patterns. Despite these advancements, the field has largely shifted toward graph-based and convolutional frameworks to improve efficiency. For instance, Social-STGCNN (Mohamed et al., 2020) represents agents as nodes in a spatio-temporal graph, using a Spatio-Temporal Graph Convolutional Neural Network (ST-GCNN) and a Time-Extrapolator CNN (TXP-CNN) to achieve a 20% improvement in Final Displacement Error (FDE) while operating 48 times faster than previous recurrent baselines.

A significant remaining hurdle is modeling non-linear motion uncertainty and the complex "multi-modal" nature of the future, where an agent may choose multiple plausible paths. To address this, generative frameworks like Social-GAN (Agrawal et al., 2025) and CVAE-based models use latent variables to produce diverse distribution samples (Jiang et al., 2025) to produce multi-modal distributions that account for human intent. Modern architectures increasingly leverage Transformer-based designs to manage these complex inputs. Models like Wayformer and Trajectron++ (Salzmann et al., 2020) demonstrate the power of fusing agent history with map features to model dependencies dynamically. Recent state-of-the-art benchmarks like Argoverse 2 (Wilson et al., 2023) and Waymo Open Motion have

catalyzed this research by providing large-scale datasets that include high-definition map context, forcing models to integrate heterogeneous scene semantics like sidewalks and lane geometry to maintain physically plausible forecasts.

In summary, pedestrian trajectory prediction research covers two persistent challenges: modeling the interaction between agents and capturing the multimodal nature of the future motion of pedestrians and vehicles. Recurrent models like LSTMs encode temporal dependencies, but their formulations are deterministic, so they often produce trajectories that fail to accurately represent the diversity of plausible futures. Social interaction models, including Social LSTM, improve behavioral realism by incorporating neighboring agents, but they typically remain deterministic and therefore are limited in expressing uncertainty. Generative approaches based on latent variable modeling, such as Conditional Variational Autoencoders (CVAEs), offer a principled framework for learning distributions over future trajectories. By introducing a stochastic latent space conditioned on observed motion, CVAEs enable the generation of multiple plausible predictions that better reflect real-world human behavior.

3 PROBLEM DECOMPOSITION & DETAILED TECHNICAL APPROACH

The problem of pedestrian trajectory prediction lends itself well to deep learning methods since the input and output formats are clear and consistent. Specifically, the datasets we use are the ETH/Hotel (Pellegrini et al., 2009) and UCY (Lerner et al., 2007) datasets, which provide human trajectory data including frame ID, pedestrian ID, X position, and Y position. The outputs are the future predicted X and Y coordinates for each pedestrian. Therefore, we can formulate this as a deep learning problem as the inputs are sequences of 2D coordinates of the pedestrians, and the model outputs sequences representing the future positions of pedestrians.

3.1 PROPOSED EXTENSIONS

Our proposed approach extends recurrent neural network-based trajectory models by incorporating Conditional Variational Autoencoders (CVAEs) to model multimodal future behaviors. We use two baseline architectures: a vanilla LSTM and Social LSTM (Alahi et al., 2016). The vanilla LSTM models each pedestrian independently, while Social LSTM incorporates human-human interactions through a social pooling mechanism. We build upon these models by integrating a CVAE framework, introducing a latent variable z to capture the inherent uncertainty and multimodality in human motion. Specifically, the model learns a conditional distribution $p(\mathbf{y} \mid \mathbf{x}, z)$, where $\mathbf{x}$ represents observed trajectories and $\mathbf{y}$ represents future trajectories. During training, a recognition network approximates the posterior $q(z \mid \mathbf{x}, \mathbf{y})$, and the model is trained using a combination of reconstruction loss and KL divergence. At inference time, samples from the learned prior $p(z \mid \mathbf{x})$ enable the generation of diverse future trajectories corresponding to different plausible human intents.

We hypothesize that incorporating CVAEs will improve performance by enabling the model to capture multimodal trajectory distributions rather than producing a single deterministic prediction (Jiang et al., 2025). Additionally, we aim to investigate whether the relative improvement from CVAEs is greater for the vanilla LSTM (which lacks interaction modeling) or for the Social LSTM (which already incorporates social context).

3.2 BASELINE TRAINING

We have found repositories that have code for both of our baseline implementations (Kothari et al., 2021), and we also found a repository containing ETH/UCY data, so we have already completed runs for our baselines on our datasets. We included 5 scenes, namely the Hotel scene from the ETH dataset, and the Zara1, Zara3, University1, and University3 scenes from the UCY dataset. We also found the ETH scene from the ETH dataset, and the Zara2 and University2 scenes from the UCY dataset. We also found scenes from the Stanford Drone Dataset (Robicquet et al., 2016), and we plan to extend our work to these datasets if time permits. Our current dataset contains a little over 10K scenes modeling pedestrian movements. We create train/test/validation splits of 0.6/0.2/0.2, yielding slightly under 8000 training examples in our dataset. We ran our vanilla LSTM and Social LSTM models for 15 epochs each, with a 0.001 learning rate at the beginning of training that eventually drops to 0.0001 based on a scheduler with step size 10, and a batch size of 32. We noticed that the Social LSTM tended to overfit quickly on the plain dataset, so for our runs on Social LSTM, we included rotation augmentation on the data. We have not experimented with hyperparameter searches yet to find optimal training parameters, so we plan to optimize and standardize settings to ensure fair comparisons between all of our models as we move forward. Furthermore, we will run our models with and without rotation augmentation applied to the input to measure the improvements from rotation augmentation. We use a prediction loss function (Gaussian NLL loss) during training, so the model learns a probabilistic trajectory distribution and tries to maximize the probability of the true trajectory. We trained both models using a single CPU, since they

are both rather lightweight. Vanilla LSTM training was completed in under 30 minutes, whereas Social LSTM took nearly 2 hours. We plan to utilize GPUs on either Colab or PACE-ICE as we scale our models up by adding CVAEs.

3.3 EVALUATION PROCEDURE

As mentioned previously, we use splits of 0.6/0.2/0.2 for train/test/val. For evaluation, we use four different metrics: ADE, FDE, Col1, and Col2. ADE represents the average L2 error between the predicted trajectory and the ground-truth trajectory over the course of the entire trajectory. FDE represents the L2 error between the predicted trajectory and the ground-truth trajectory at the final timestep. Col1 represents the collision rates between pedestrians based on the model’s predicted trajectories. If the model makes predictions for two different pedestrians and those trajectories run into one another, then that would count as a Col1 collision. Col2 represents the collision rates between pedestrians comparing model-predicted trajectories to ground-truth trajectories. These four metrics are commonly used to benchmark performance in pedestrian trajectory models, and they provide a comprehensive evaluation of accuracy and social consistency. The formulas for ADE and FDE are as follows:

$$ADE = \frac{1}{T} \sum_{t=1}^T \|p_t - \hat{p}_t\|$$

$$FDE = \|p_T - \hat{p}_T\|$$

where p represents the ground-truth trajectory, $\hat{p}$ represents the predicted trajectory, and t represents the timestep of the trajectory.

3.4 RESEARCH HYPOTHESES

We propose three research hypotheses (H1-H3):

- Applying rotation augmentation to the trajectory coordinates during training will improve test-time trajectory predictions compared to training without rotation augmentation, with all other training parameters kept constant
- Adding a CVAE component to both the vanilla LSTM and Social LSTM baselines will improve trajectory forecasting performance by better modeling multimodal future behavior, leading to lower ADE and FDE and potentially lower collision rates.
- Adding a CVAE component to Social LSTM will provide a larger performance gain than adding a CVAE component to vanilla LSTM, since the latent variable can model uncertainty in social interaction-based future behavior, whereas vanilla LSTM does not explicitly represent pedestrian interactions.

4 EXPERIMENTS & RESULTS

We evaluate our baseline models on the ETH/UCY trajectory forecasting benchmark described in the previous section. Our goal at the milestone stage is to establish a reliable deterministic baseline before extending the models with a CVAE for multimodal prediction. In particular, we compare a vanilla LSTM, which predicts each pedestrian independently, against a Social LSTM, which augments the recurrent model with social pooling so that neighboring pedestrians can influence the forecast.

4.1 EXPERIMENTAL SETUP

We use the same five-scene subset described earlier. The dataset contains a little over 10K trajectory scenes, which we split into train/validation/test partitions using a 0.6/0.2/0.2 ratio. For both baselines, we observe 9 time steps and predict the next 12 time steps. Training uses a batch size of 32, an initial learning rate of 0.001, and a step scheduler that reduces the learning rate after epoch 10. Both models are trained with Gaussian prediction loss so that the decoder learns a probabilistic trajectory distribution rather than only a point estimate.

The Social LSTM was more prone to overfitting on the unaugmented data, so we trained that model with rotation augmentation. In contrast, the vanilla LSTM was trained without social context and without augmentation. Both models were trained on CPU. In practice, the vanilla LSTM completed training in under 30 minutes, while the Social LSTM required nearly two hours, reflecting the additional computational cost of explicit interaction modeling.

4.2 EVALUATION METRICS

We report four standard trajectory forecasting metrics: ADE, FDE, Col1, and Col2. ADE measures the average Euclidean distance between the predicted and ground-truth trajectories over the full prediction horizon, while FDE measures the Euclidean distance at the final predicted time step. Col1 measures collisions between predicted pedestrian trajectories, and Col2 measures collisions between predicted trajectories and surrounding ground-truth pedestrian trajectories. Together, these metrics capture both pointwise accuracy and social plausibility.

4.3 OVERALL QUANTITATIVE RESULTS

Table 1 summarizes the overall performance of the two baselines on 1,556 evaluation trajectories.

Model	ADE	FDE	Col1	Col2	No.
Vanilla LSTM	0.53	1.17	13.30	15.55	1556
Social LSTM	0.54	1.17	8.80	15.75	1556

Table 1: Overall baseline performance on the held-out evaluation set. Lower is better for all metrics.

We also include the full evaluator output in Section 6, which provides a finer-grained breakdown across trajectory categories and interaction sub-types.

4.4 METHOD IMPLEMENTATION PROGRESS

While full CVAE results are not yet available at the milestone stage, we have made concrete progress toward our proposed extensions. We have split into two subgroups: one focused on implementing the VAE encoder (the recognition network $q(z \mid \mathbf{x}, \mathbf{y})$) and one on the VAE decoder (the generative network $p(\mathbf{y} \mid \mathbf{x}, z)$). Both subgroups are drawing on existing CVAE implementations as references to guide the architecture design. The planned integration attaches the CVAE on top of the existing LSTM encoder outputs, with the latent variable z concatenated into the decoder input at each step. We expect to complete initial CVAE training runs on top of the vanilla LSTM baseline by April 10–15, as outlined in our timeline below.

4.5 ANALYSIS

The two models exhibit a clear tradeoff. The vanilla LSTM achieves the best displacement accuracy, with a slightly lower ADE of 0.53 compared to 0.54 for the Social LSTM, while both models obtain the same FDE of 1.17. However, the Social LSTM produces substantially fewer predicted collisions, reducing Col1 from 13.30 to 8.80. This is a relative reduction of roughly 34%, which suggests that explicitly modeling nearby pedestrians improves the realism and safety of the predicted trajectories even when the improvement is not reflected in ADE or FDE.

The Col2 metric remains nearly unchanged between the two models, with the vanilla LSTM at 15.55 and the Social LSTM at 15.75. This indicates that the social model is better at preventing conflicts among jointly predicted trajectories, but it does not yet fully solve the broader challenge of matching the motion of all surrounding real pedestrians. In other words, social pooling helps enforce consistency within the forecasted multi-agent scene, but there is still room to improve robustness to the full uncertainty of human motion.

The category-level breakdown in Section 6 further suggests that the benefit of social interaction modeling depends on scene structure. Some trajectory categories favor slightly better displacement accuracy from the vanilla LSTM, while more interactive scenes benefit from the socially aware model’s lower collision rate. This reinforces the importance of evaluating with both geometric accuracy metrics and interaction-sensitive metrics rather than relying on ADE and FDE alone.

4.6 IMPLICATIONS FOR THE FINAL PROJECT

These results support the motivation for the next stage of the project. First, they confirm that interaction modeling is valuable even before introducing multimodality, because the Social LSTM yields more socially plausible forecasts. Second, they also show the main limitation of deterministic models: despite improved collision behavior, both baselines still collapse to a single forecast and achieve nearly identical FDE. This is precisely the setting where a CVAE

extension is promising. By introducing a latent variable to model multiple plausible futures, we expect to improve not only trajectory accuracy but also the ability to represent uncertainty in ambiguous scenes.

At the same time, this milestone exposes an important experimental limitation. The current comparison is informative, but it is not yet a fully controlled ablation, since the Social LSTM was trained with rotation augmentation while the vanilla LSTM was not. As part of the final phase of the project, we will standardize augmentation and training budgets across all models so that we can isolate the contribution of interaction modeling, multimodal latent variables, and their combination more rigorously.

5 NEXT STEPS

Our current comparison is informative, but it is not a perfectly controlled ablation. The two runs differ not only in architecture, but also in training settings: the vanilla model ran without any rotation augmentation, while the social model ran with rotation augmentation. Furthermore, we did not modify the number of layers or embedding dimensions of the Social LSTM, so we need to rerun the Social LSTM baseline in accordance to the model settings presented in the Social LSTM paper (Alahi et al., 2016). Thus, we have currently presented working baselines and a training pipeline, but we must rerun our baselines to ensure the most fair and valid results.

The next steps are straightforward:

- rerun both models under the same augmentation policy
- visualize failure cases from `test_pred` to determine where the social model still collides with ground-truth neighbor motion
- add our CVAE implementations on top of our baselines and evaluate results

Our tentative timeline is:

- April 3: rerun baselines with matched augmentation and training settings;
- April 10–15: add the CVAE component to the vanilla LSTM and evaluate results;
- April 20–25: add the CVAE component to the Social LSTM and evaluate results;
- Late April–end of term: analyze results, compare all four models, and complete the final report.

One potential challenge is ensuring that performance differences are actually caused by model design rather than by inconsistent training choices, so we will standardize augmentation, hyperparameters, and evaluation settings across all runs. A second challenge is that CVAE training can be unstable or may not improve deterministic metrics such as ADE and FDE as much as expected. To address this, we will evaluate not only aggregate metrics but also qualitative trajectory samples and failure cases, especially in ambiguous or highly interactive scenes.

Our criterion for final success is not limited to showing that the CVAE models outperform the baselines on ADE, FDE, Col1, and Col2, although that is our primary hypothesis. More broadly, success means producing a careful comparison across all four models and being able to explain the observed trends in a meaningful way. If the CVAE extensions improve performance, we will analyze where those gains come from and whether they are larger for the socially aware model. If they do not improve performance, we will still consider the project successful if we can identify plausible reasons for the outcome, such as training instability, insufficient latent capacity, or limitations in the dataset and evaluation setup. This will allow us to write a strong final report centered not just on results, but on a clear and thoughtful analysis of what worked, what did not, and why.

REFERENCES

- Subham Agrawal, Nico Ostermann-Myrau, Nils Dengler, and Maren Bennewitz. Peroi: A pedestrian-robot interaction dataset for learning avoidance, neutrality, and attraction behaviors in social navigation, 2025. URL <https://arxiv.org/abs/2503.16481>.
- Alexandre Alahi, Kratarth Goel, Vignesh Ramanathan, Alexandre Robicquet, Li Fei-Fei, and Silvio Savarese. Social lstm: Human trajectory prediction in crowded spaces. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016.
- J. Jiang, K. Yan, X. Xia, and B. Yang. A survey of deep learning-based pedestrian trajectory prediction: Challenges and solutions. *Sensors*, 25(3):957, February 2025. doi: 10.3390/s25030957.

- Parth Kothari, Sven Kreiss, and Alexandre Alahi. Human trajectory forecasting in crowds: A deep learning perspective. *IEEE Transactions on Intelligent Transportation Systems*, pp. 1–15, 2021. doi: 10.1109/TITS.2021.3069362.
- Alon Lerner, Yiorgos Chrysanthou, and Dani Lischinski. Crowds by example. *Computer Graphics Forum*, 26, 2007. URL <https://api.semanticscholar.org/CorpusID:17374844>.
- A. Mohamed, K. Qian, M. Elhoseiny, and C. Claudel. Social-stgcnn: A social spatio-temporal graph convolutional neural network for human trajectory prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14412–14420, June 2020. doi: 10.1109/cvpr42600.2020.01443.
- S. Pellegrini, A. Ess, K. Schindler, and L. van Gool. You’ll never walk alone: Modeling social behavior for multi-target tracking. In *2009 IEEE 12th International Conference on Computer Vision*, pp. 261–268, 2009. doi: 10.1109/ICCV.2009.5459260.
- A. Robicquet, A. Sadeghian, A. Alahi, and S. Savarese. Learning social etiquette: Human trajectory understanding in crowded scenes. In *Lecture Notes in Computer Science*, pp. 549–565, 2016. doi: 10.1007/978-3-319-46484-8_33.
- Tim Salzmann, Boris Ivanovic, Punarjay Chakravarty, and Marco Pavone. Trajectron++: Dynamically-feasible trajectory forecasting with heterogeneous data. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 683–700. Springer, 2020.
- B. Wilson et al. Argoverse 2: Next generation datasets for self-driving perception and forecasting. *arXiv preprint arXiv:2301.00493*, 2023. URL <https://arxiv.org/abs/2301.00493>. accessed Feb. 13, 2026.
- Z. Zhang et al. Stinet: Spatio-temporal-interactive network for pedestrian detection and trajectory prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11343–11352, June 2020. doi: 10.1109/cvpr42600.2020.01136.

6 APPENDIX

Type	Sub-Type	Model	No.	ADE	FDE	Col I	Col II	Top3 ADE	Top3 FDE	NLL
I	–	lstm_social	41	0.30	0.68	17.07	14.63	0.00	0.00	0.00
I	–	lstm_vanilla	41	0.33	0.78	9.76	14.63	0.00	0.00	0.00
II	–	lstm_social	153	0.40	0.81	10.46	20.26	0.00	0.00	0.00
II	–	lstm_vanilla	153	0.41	0.84	16.99	16.99	0.00	0.00	0.00
III	–	lstm_social	1166	0.56	1.22	9.61	15.87	0.00	0.00	0.00
III	–	lstm_vanilla	1166	0.56	1.22	14.41	16.72	0.00	0.00	0.00
III	LF	lstm_social	148	0.55	1.19	12.16	13.51	0.00	0.00	0.00
III	LF	lstm_vanilla	148	0.51	1.10	14.19	20.27	0.00	0.00	0.00
III	CA	lstm_social	395	0.59	1.28	15.95	17.47	0.00	0.00	0.00
III	CA	lstm_vanilla	395	0.59	1.28	19.49	20.00	0.00	0.00	0.00
III	Grp	lstm_social	380	0.50	1.12	9.74	18.68	0.00	0.00	0.00
III	Grp	lstm_vanilla	380	0.49	1.09	15.00	20.53	0.00	0.00	0.00
III	Oth	lstm_social	445	0.58	1.26	6.29	14.83	0.00	0.00	0.00
III	Oth	lstm_vanilla	445	0.58	1.28	11.69	13.26	0.00	0.00	0.00
IV	–	lstm_social	196	0.56	1.24	1.02	11.73	0.00	0.00	0.00
IV	–	lstm_vanilla	196	0.54	1.20	4.59	7.65	0.00	0.00	0.00

Table 2: Comprehensive Model Performance Breakdown Across Workload Types and Sub-Types